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A. RAG Ablation Study

As mentioned in Section 5, we compared various values of $K \in \{0, \dots, 5\}$, shown in Table 3. We were unable to find an optimal value of K per model, per 0/5-shot, or per task. In fact, other than Anatomy that worked well with $K = 2$ consistently, there seems to be no patterns that aid in predicting the performance per K , unlike the results presented in (Lewis et al., 2020) for other setups. Moreover, the gap between the best and worst performing K s can be large.

Unfortunately, we must conclude that this additional hyperparameter is unstable. This is a downside of using RAG in practice, and the choice of K cannot be ignored.

Task	Model	# Retrieved documents (k)				
		1	2	3	4	5
Anatomy (0-shot)	Mistral 7B	0.615	0.681	0.630	0.644	0.622
	Llama2 7B	0.444	0.489	0.467	0.474	0.481
	Orca2 7B	0.607	0.637	0.600	0.585	0.637
Anatomy (5-shot)	Mistral 7B	0.659	0.667	0.659	0.681	0.674
	Llama2 7B	0.496	0.563	0.541	0.526	0.526
	Orca2 7B	0.630	0.659	0.600	0.600	0.600
Astronomy (0-shot)	Mistral 7B	0.651	0.678	0.678	0.664	0.664
	Llama2 7B	0.447	0.434	0.447	0.434	0.467
	Orca2 7B	0.711	0.730	0.730	0.750	0.730
Astronomy (5-shot)	Mistral 7B	0.704	0.684	0.658	0.684	0.724
	Llama2 7B	0.461	0.447	0.474	0.428	0.454
	Orca2 7B	0.730	0.737	0.750	0.743	0.763
Biology (0-shot)	Mistral 7B	0.736	0.722	0.757	0.743	0.736
	Llama2 7B	0.438	0.472	0.493	0.479	0.472
	Orca2 7B	0.639	0.618	0.639	0.625	0.639
Biology (5-shot)	Mistral 7B	0.722	0.778	0.778	0.771	0.743
	Llama2 7B	0.500	0.521	0.507	0.465	0.472
	Orca2 7B	0.625	0.639	0.625	0.660	0.660
Chemistry (0-shot)	Mistral 7B	0.450	0.470	0.470	0.500	0.470
	Llama2 7B	0.320	0.320	0.300	0.380	0.360
	Orca2 7B	0.370	0.420	0.400	0.410	0.440
Chemistry (5-shot)	Mistral 7B	0.540	0.490	0.500	0.510	0.470
	Llama2 7B	0.280	0.320	0.340	0.340	0.380
	Orca2 7B	0.390	0.430	0.400	0.430	0.470
Prehistory (0-shot)	Mistral 7B	0.728	0.725	0.750	0.735	0.728
	Llama2 7B	0.481	0.460	0.457	0.457	0.429
	Orca2 7B	0.648	0.645	0.660	0.670	0.679
Prehistory (5-shot)	Mistral 7B	0.710	0.750	0.759	0.756	0.762
	Llama2 7B	0.512	0.485	0.525	0.519	0.531
	Orca2 7B	0.660	0.688	0.685	0.698	0.688

Table 3. RAG ablation study.

B. Paraphrase Examples

Below is the prompt we used to generate paraphrases with GPT-4: